

# Chapter 17.0

## Object Detection Algos

தமிழில்

Less than 15 Mins

**AI COACH JOHN**

CEO & FOUNDER

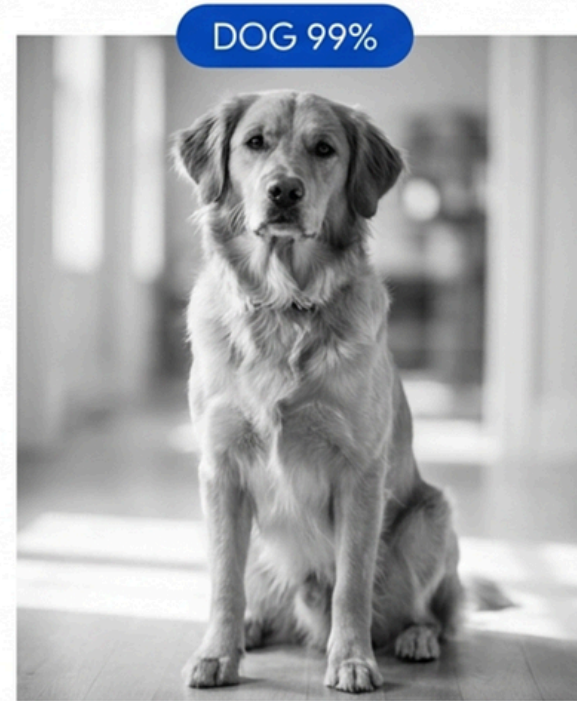
# Agenda at a glance

1. What is Object Detection? Right Definition
2. Why Object Detection is hard?
3. *3 Fundamental Problem* with Object Classification
4. {.....}

## Point 1

*What is **Object Detection**? Right Definition*

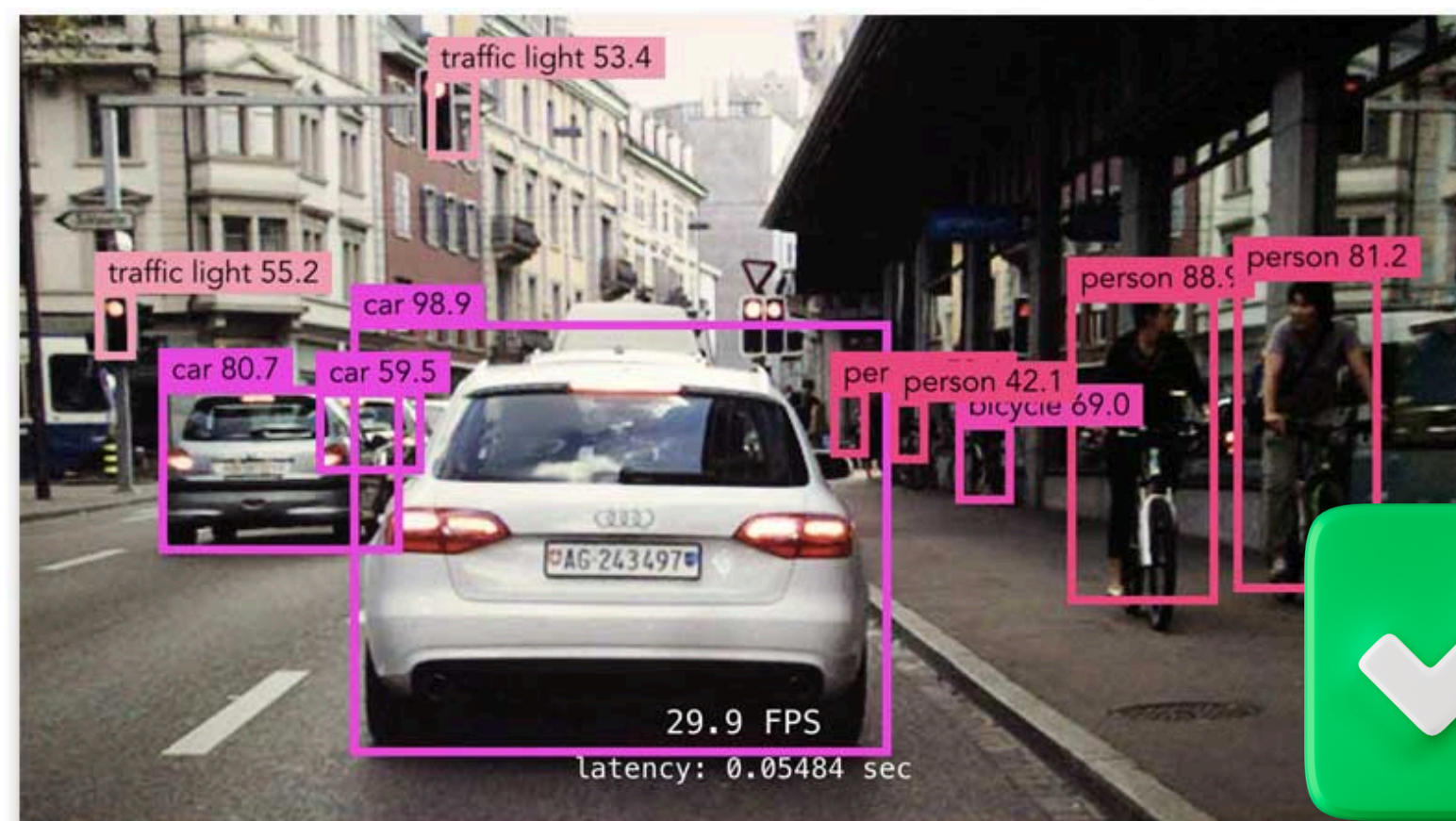
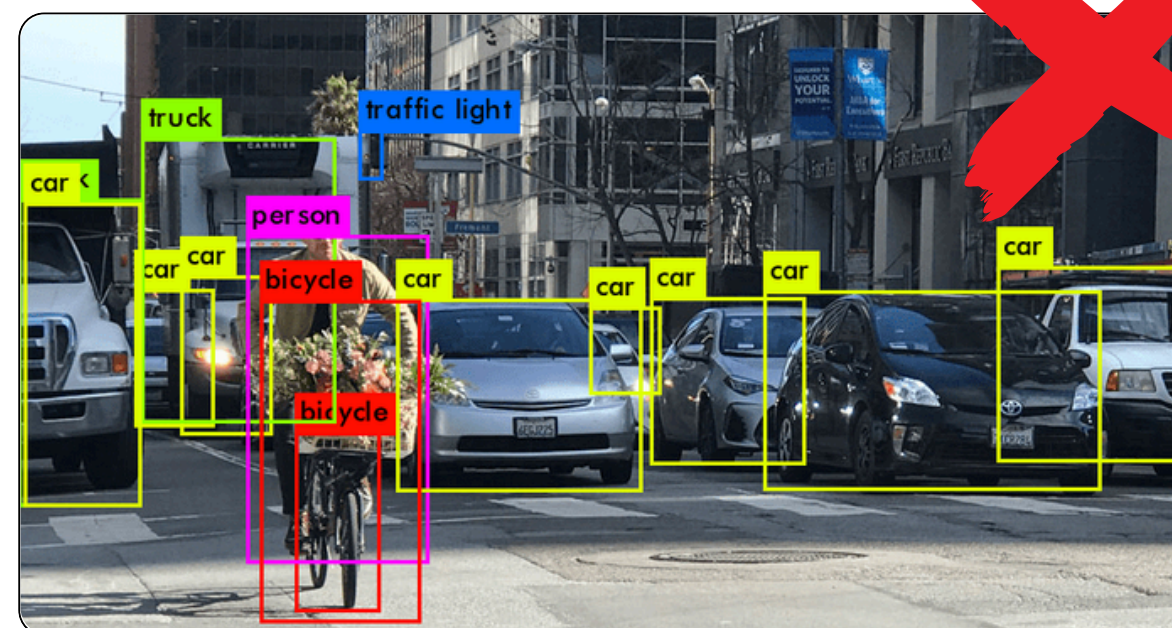
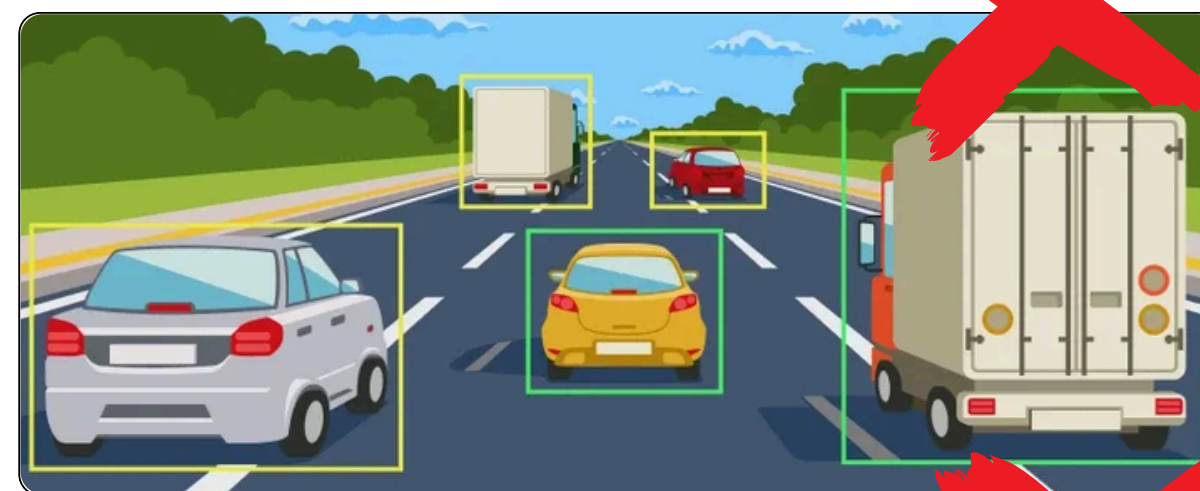
# What is Object Detection?



What is it? (Classification)



Where is it? (Localization)



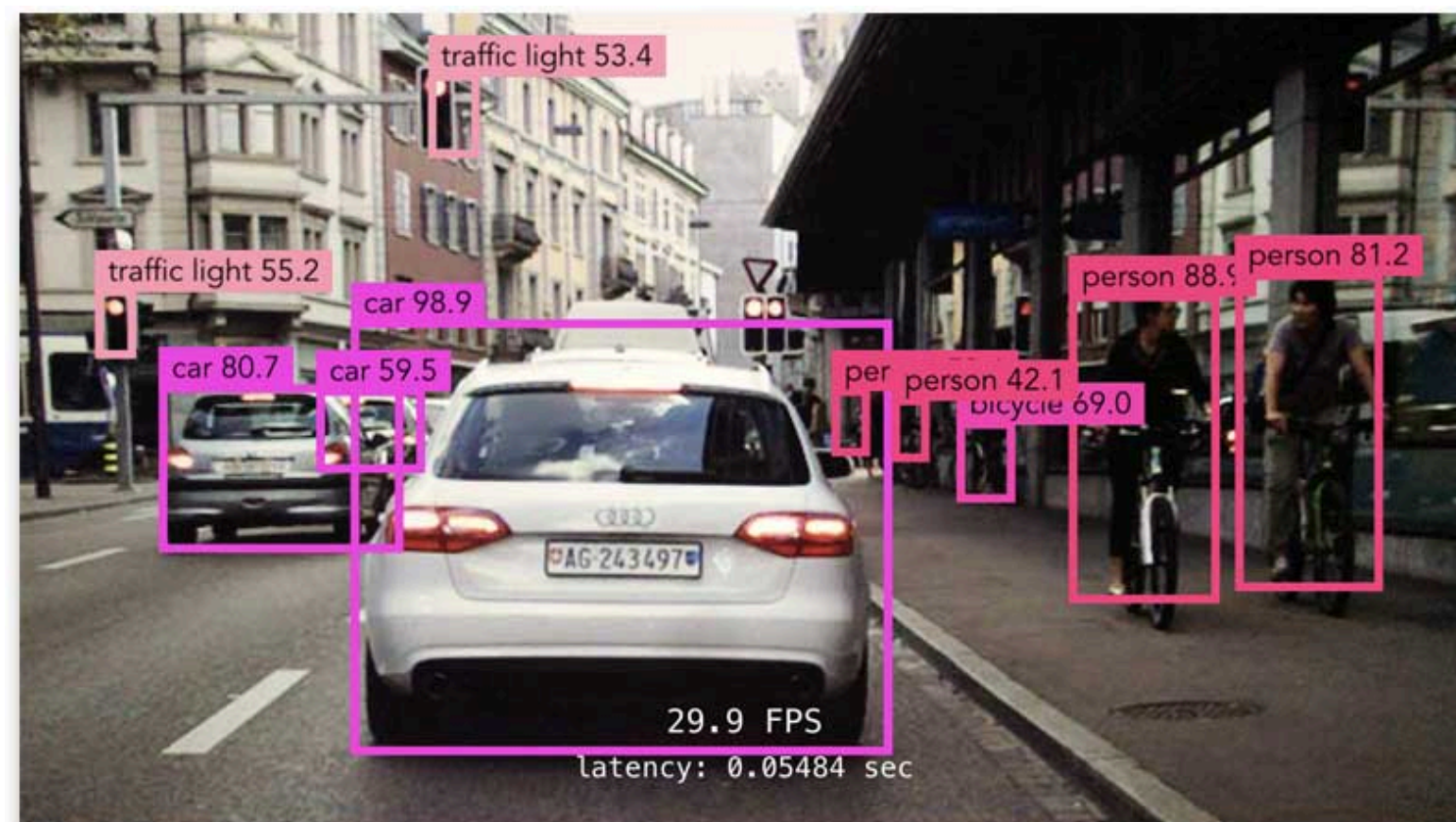
## Object Detection

1. **Localization** (Bounding Boxes)
2. **Classification**
3. **Confidence**

## Point 2

*Why Object Detection is hard?*

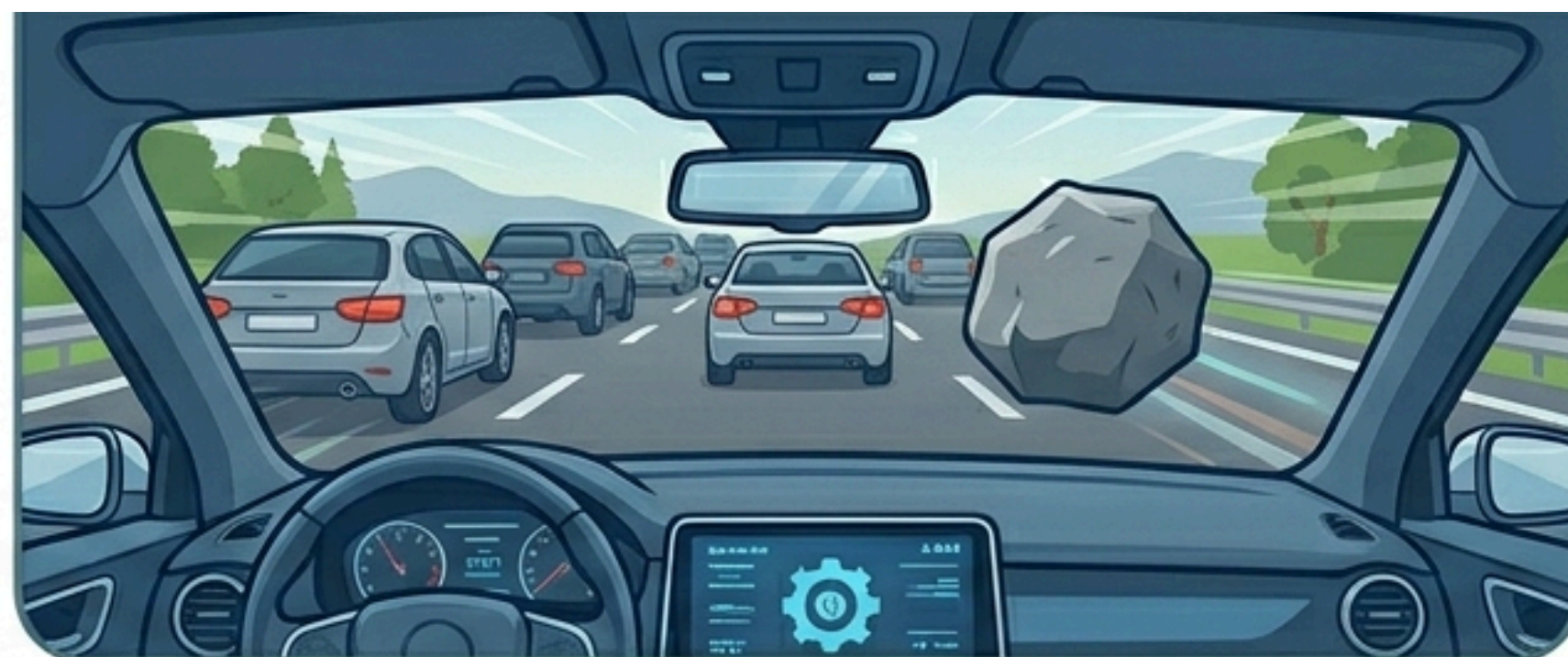
# Why *Object Detection* is Hard?



*In Real-time*

**Speed matters**

*This is the REAL CHALLENGE.*



## Point 3

***3 Fundamental Problem with Object Classification***

# 3 Fundamental Problem with Object Classification

**Problem 1** → 1 Frame at a Time

**Problem 2** → 1 Object at a Time

**Problem 3** → Not **Real-Time** Ready



Multiple Frames per second  
Good if you can process **Min 30 FPS** – **Max 60 FPS**



So, you need to identify where the particular object is  
in a **Motion Videos** and perform 3 tasks.



# WHAT'S NEXT?

*In 2014, Ross Girshick & his team, they approached this problem with a model called **R-CNN**.*

# Summary

1. What is Object Detection? Right Definition
2. Why Object Detection is hard?
3. *3 Fundamental Problem* with Object Classification
4. Who was recognized and for what?